

Intended Use

This project explores the use of linear probes on hidden representations from a frozen language model to detect potentially toxic user prompts before a model generates a response.

The intended use is research into AI safety, trustworthiness, and mechanistic interpretability. The approach may contribute to lightweight guardrail systems that identify potentially harmful inputs using information already encoded within a model's internal representations.

Potential Benefits

A successful toxicity probe could support earlier detection of harmful or undesirable prompts before they reach later stages of a language model.

Potential benefits include:

Improved AI safety by identifying potentially toxic inputs.

Lightweight detection, since the language model remains frozen and only a linear classifier is trained.

Improved understanding of where toxicity-related information emerges within model representations.

Potential use as an additional layer within a broader safety or moderation system.

Potential Risks

The system may incorrectly classify benign content as toxic, particularly when text contains identity terms, profanity used in a non-harmful context, sarcasm, quotations, or discussions about harmful language.

False negatives are also possible, allowing toxic or harmful content to pass through undetected.

The datasets used for training may contain biases related to language, culture, demographic groups, writing styles, or the contexts in which particular terms are considered harmful.

These biases may be reflected in the trained probe.

Because toxicity is context-dependent and subjective, a binary classification system may oversimplify certain forms of language.

Fairness and Bias

The project uses multiple toxicity datasets to improve diversity in the training data and reduce reliance on a single source or labeling standard.

However, combining datasets does not eliminate bias. Different datasets may define or annotate toxicity differently, and the resulting classifier may perform differently across demographic groups, languages, dialects, and communication styles.

Special attention should be given to false positives involving identity-related language and to differences in performance across different populations.

Privacy

The project uses publicly available research datasets and does not collect new personal data.

The goal of the project is not to identify individuals or infer personal characteristics. Text is used solely for toxicity classification and representation analysis.

Transparency and Accountability

The model is based on a frozen Gemma language model and a logistic regression probe trained on mean-pooled hidden-state representations.

The use of a linear probe provides a relatively simple and interpretable classification layer compared with a larger end-to-end classifier. However, the internal language-model representations themselves remain complex and should not be interpreted as directly revealing why an individual input is classified as toxic.

The system should not be used as the sole decision-maker in high-stakes moderation or enforcement settings without additional evaluation, monitoring, and human oversight.

Limitations

Performance on the competition evaluation set does not guarantee equivalent performance in real-world deployment.

The probe was trained on English-language toxicity datasets and may not generalize equally well to other languages or domains.

The binary safe/toxic formulation does not capture the severity, target, context, or type of harmful language.

The selected training data and decision threshold may influence the balance between false positives and false negatives.

This work should therefore be considered a research prototype rather than a complete standalone toxicity moderation system.